

PERSONALIZED LINUX & GIT CHEAT SHEET

Quick student reference • A4 • commands + examples + three-tree workflow

1. BASIC LINUX COMMANDS			
Command	Sample usage	What it does	
pwd	pwd	Show current folder/path.	
ls	ls -la	List files; -l details, -a hidden files.	
cd	cd Documents	Move to another directory.	
mkdir	mkdir project	Create a new directory.	
touch	touch notes.txt	Create an empty file/update timestamp.	
cp	cp a.txt backup.txt	Copy a file/folder.	
mv	mv old.txt new.txt	Move or rename.	
rm	rm notes.txt	Remove a file; use -r for directories.	
cat	cat notes.txt	Display file contents.	
grep	grep "error" log.txt	Search text matching a pattern.	
chmod	chmod +x script.sh	Change file permissions.	
clear	clear	Clear the terminal screen.	
2. BASIC GIT COMMANDS			
Command	Sample usage	Description (own wording)	Common options
git init	git init	Create a new local repository.	-q, -b
git status	git status	Show changed/staged files.	-s, -b
git add	git add .	Put changes into staging area.	-A, -p, -n
git commit	git commit -m "Update"	Save staged changes as a snapshot.	-m, -a, --amend
git log	git log --oneline	View commit history.	--oneline, --graph, -n
git diff	git diff	Compare changes before commit.	--staged, --stat, --check
git restore	git restore file.txt	Restore file content from a Git version.	--staged, --source
git branch	git branch	List/create/delete branches.	-a, -d, -m
git switch	git switch main	Switch to another branch.	-c, -d
git checkout	git checkout main	Older multi-purpose branch/file switch command.	-b, -f
git merge	git merge feature	Combine another branch into current branch.	--no-ff, --abort
git tag	git tag v1.0	Mark a specific commit/version.	-a, -d, -l
git clone	git clone URL	Copy a remote repository locally.	-b, --depth, --single-branch
3. GIT THREE-TREE ARCHITECTURE Working Directory → Index / Staging Area → HEAD / Repository Edit files → git add → git commit • Working Directory: files you are currently editing. • Index/Staging Area: changes prepared for the next commit. • HEAD: points to the current branch/last committed snapshot. Quick flow: edit → add → commit → history		REFERENCE • Git Documentation / Reference: git-scm.com/docs • Pro Git, "Reset Demystified" (Three Trees): git-scm.com/book/en/v2/Git-Tools-Reset-Demystified • GNU Coreutils Manual: gnu.org/software/coreutils/manual/ • Git command details were checked against official Git documentation.	

AI USE DECLARATION: I used ChatGPT (GPT-5.6 Luna) to help research, organize, and format the cheat sheet. I reviewed and edited the content and take responsibility for the final work.